



**Politecnico
di Torino**



Presented by Group - G

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OVERVIEW

- Objectives
- Architecture Recap
- Test Results
- WBS
- Gantt Chart

OBJECTIVES

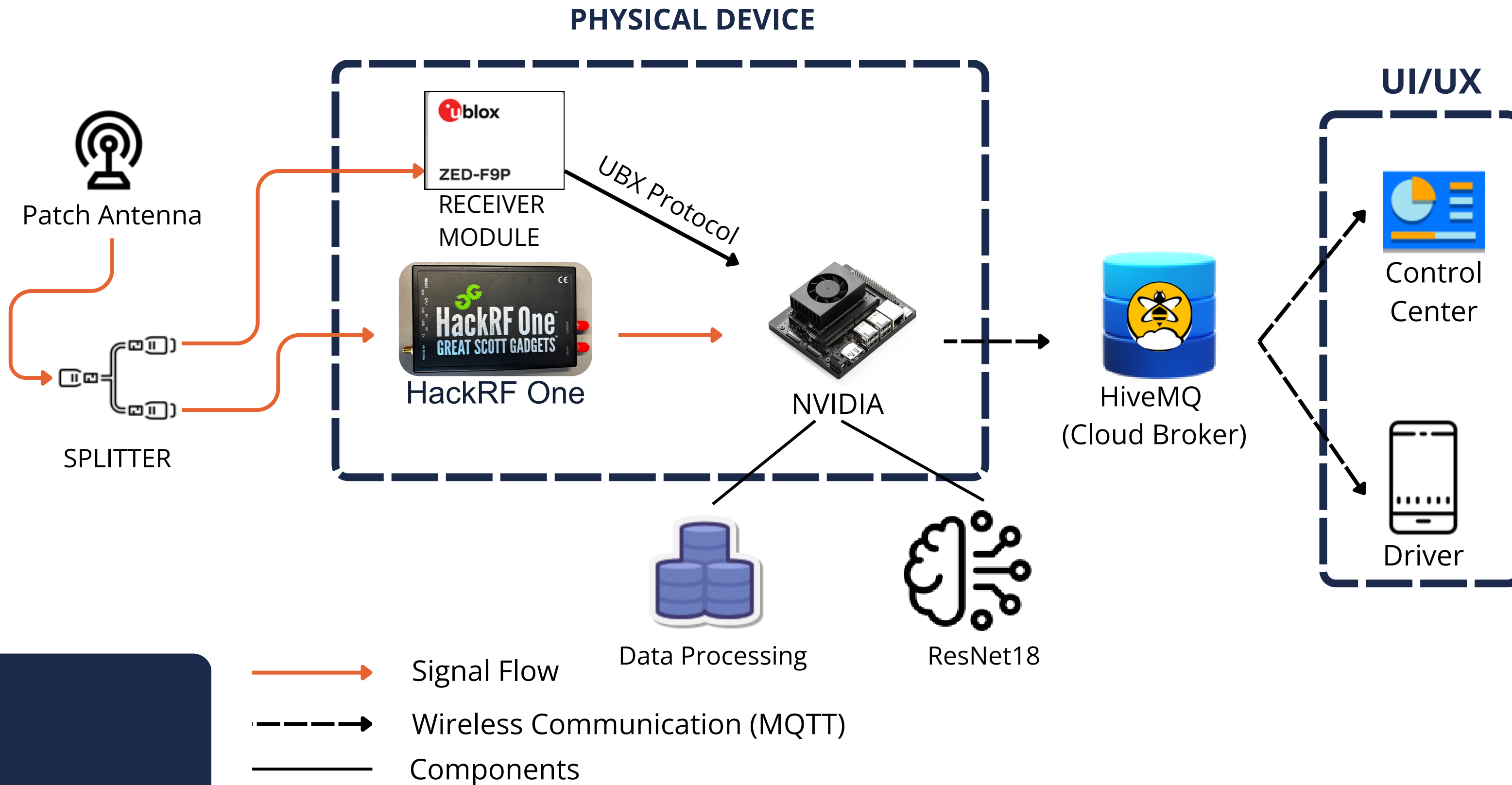
The Problem: The Hidden Vulnerability of GNSS

- Modern society critically dependent on GNSS for different purposes, malicious radio devices known as “jammers” are cheap, easy to acquire, and can blind GNSS receivers
- Real-world impact: **Organized cargo theft** (e.g., Gucci case study)
 - Thieves jam tracking systems to blind fleet monitoring
 - High-value goods stolen while systems are compromised
 - Professional monitoring solutions too expensive for wide deployment

The solution: An Intelligent and Autonomous Guardian

- Small, cheap monitoring station deployed on trucks that constantly monitor to the GNSS frequency bands
- Using advanced signal processing algorithms and AI, the system analyzes the radio spectrum to identify anomalies in real time
- Once a jammer is detected and classified, the system automatically notify the company of the current situation and sends an alert to a central station

ARCHITECTURE



TEST PLAN

The test plan was organized into 4 test areas, each derived directly from our requirements. Every test had a unique ID, maps to specific requirements, defines the exact procedure, and states a quantifiable pass criterion

AREA 01

Hardware
&
Signal Acquisition Tests

UREQ_1, FREQ_1,
FREQ_2, TREQ_1,
TREQ_2

AREA 02

AI Model
&
Classification Tests

FREQ_3, TREQ_3,
TREQ_4, TREQ_5

AREA 03

Communication
&
Alerting Tests

UREQ_3, FREQ_5,
TREQ_6

AREA 04

User Interface Tests

UREQ_2, FREQ_4,
TREQ_7

HARDWARE & SIGNAL ACQUISITION TESTS

Test ID	Procedure	Pass Criteria	Outcome
T1.1	Acquire IQ samples from the antenna for 10 seconds at rate 10e6, verify the output buffer is non-zero	Zero dropped samples or buffer overflows during continuous capture	PASS
T1.2	Simultaneously acquire position and raw IQ data	Successful location data stream alongside IQ snapshots in <60s	PASS
T1.3	Start the system and let it run for 1 hour without human intervention	100% uptime over the time window	PASS
T1.4	Power on the device and wait for it to reach "ACTIVE" status on the UI	System ready for monitoring in < 60 seconds from power-on	System ready for monitoring in 70s due to NVIDIA pre accension tests

SATISFIED REQUIREMENTS

UREQ_1	The user shall be able to deploy and operate the system simply by connecting power and network, without complex configuration procedures
UREQ_2	The Control Center shall have a simple, intuitive way to see if GNSS signals are clean or compromised, and what type of interference is disrupting them
UREQ_3	Automatic warning shall be delivered to the user when jamming is detected
FREQ_1	continuously monitor GNSS signals (L band) multi-constellation
FREQ_2	acquire and then process GNSS signals to extract time-frequency representations
FREQ_3	integrate Artificial Intelligence to correctly identify and classify jamming interference across multiple interference types with different power of jamming signals JSR
FREQ_4	provide the user a simple UI to visualize the output of the AI algorithm and signal status
FREQ_5	automatically send an immediate warning if jamming is present to both the UI and a control center

SATISFIED REQUIREMENTS

TREQ_1	Multi-band GNSS patch antenna suitable for vehicle-mounted applications, providing reliable signal reception in variable conditions
TREQ_2	An SDR-based RF front-end capable of capturing GNSS signals, with an ADC that provides sufficient sampling rate and resolution for high-quality GNSS signal processing
TREQ_3	computational unit supporting real time AI inference and signal processing operations
TREQ_4	signal processing pipeline to elaborate input data using STFT
TREQ_5	test environment with capability to generate synthetic jamming signals with different power of jamming signals JSR
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TREQ_7	web based UI providing real-time status updates

SysReq document [here](#)

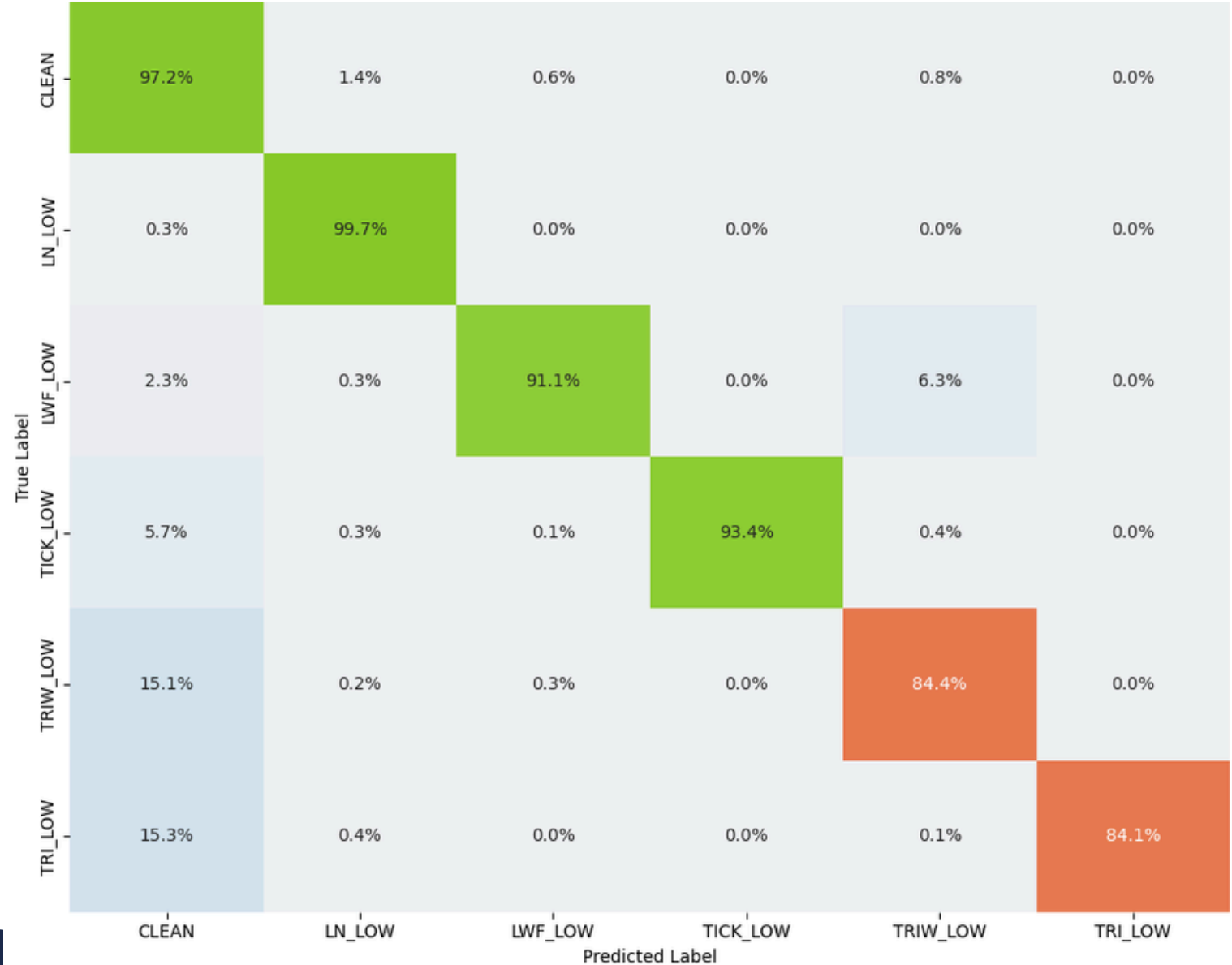
AI MODEL & CLASSIFICATION TESTS

Test ID	Procedure	Pass Criteria	Outcome
T2.1	Inject 500 clean GNSS signal snapshots and record the AI model output class	≥95% of snapshots classified as clean	PASS 97% classified correctly 3% of LOW jamming
T2.2.1	Generate jamming with HackRF at specific JSR levels (LOW, MID, HIGH) for all Jamming types (LWF, LN, TRI, TRIW, TICK), run segments through the pipeline, record outputs	Per-class accuracy ≥90%	PASS Average accuracy: 95%
T2.2.2	Generate jamming with HackRF at LOWest , for all Jamming types (LWF, LN, TRI, TRIW, TICK), run segments through the pipeline, record outputs	Per-class accuracy ≥90%	3/5 PASS
T2.3	Measure the full pipeline time over 500 snapshot	Mean task completion ≤ 100 ms	FAIL

AI MODEL & CLASSIFICATION TESTS

T2.1 to **reduce false positives**, the system publishes an alert only after **3 consecutive signal snapshots** are classified with the same jamming type so that a single anomalous reading does not trigger an alarm. This design choice protects both the driver and the fleet operator from unnecessary responses.

T2.2.2

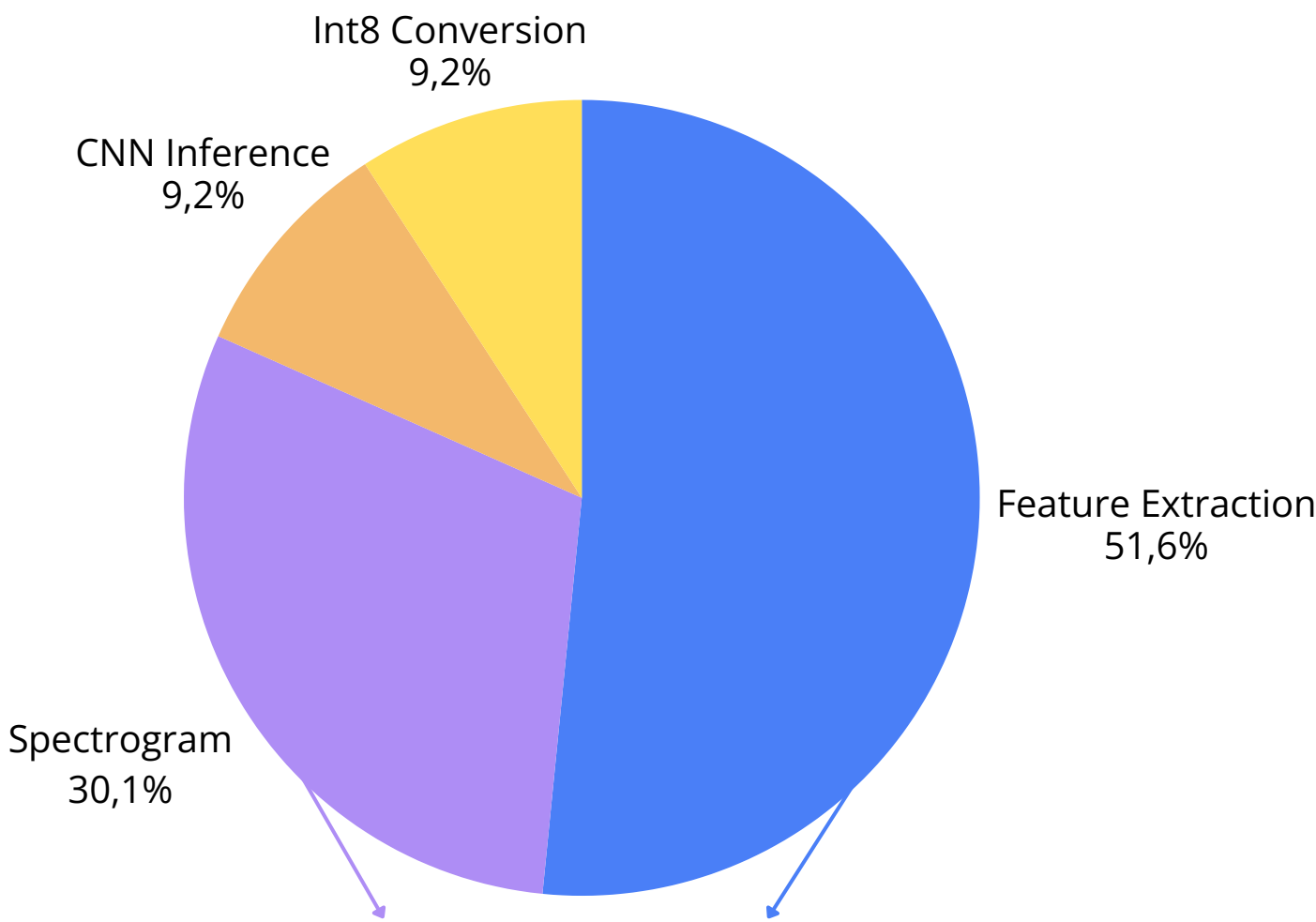


Hardest test scenario: LOW JSR to test model sensibility to False Positives

T2.3

Average pipeline time in steady state : **350ms**

Latency Breakdown



Bottleneck: ARM CPU-bound preprocessing

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COMMUNICATION & ALERTING TESTS

Test ID	Procedure	Pass Criteria	Outcome
T3.1	Trigger a simulated jamming event on the Processing Unit and measure time to UI notification	50/50 alerts delivered and mean latency ≤ 5 s	PASS
T3.2	Forcibly disconnect the device (power cut) to verify that the HiveMQ broker publishes Last Will Testament message to the Control Center and Driver UI	Control Center UI receives the LWT message and changes the truck status to "OFFLINE" in the Active Vehicles Table	PASS LWT configured to be sent in 60s to both UIs
T3.3	Simulate a brief network interruption during a jamming event and verify that the alert is re-delivered once connectivity is restored (QoS level 1 guarantee)	Alert is queued and delivered to Control Center after reconnection without user action in 50/50 tests.	PASS
T3.4	Connect two simulated devices with different vehicle plate IDs and trigger jamming only on 1 device and verify that the Control Center receives the alert only on the correct topic	0 messages on the wrong topic in 10 test events	PASS

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USER INTERFACE TESTS

Test ID	Procedure	Pass Criteria	Outcome
T4.1	Open Control Center UI and the Driver mobile UI on separate devices and measure load time from first network request to fully rendered page	Load time $\leq 5s$ on both desktop and mobile browser in 50/50 tests	Driver UI load time $<5s$ Control Center load time $<10s$
T4.2	With the UI open, trigger a jamming event and measure the time from MQTT message publication to the moment the jamming counter increments and the pop-up alert appears in the browser	UI update latency $\leq 5s$ in 10/10 tests	PASS
T4.3	Trigger 5 distinct jamming events with different jamming types and verify that all 5 appear in the jamming event table with correct Plate, Jamming Type, Time, and Position fields.	All 5 rows populated correctly with no missing or incorrect fields	PASS
T4.4	Populate Alert Table and apply filters for all possible parameters on the dashboard	Only events matching the filter criteria are displayed both in the Alert Table and Map in 10 filter test cases	PASS

USER INTERFACE LAYOUT

Added Features from CDR:

- Clear History button removes all entries in the Alert Table from the current session log and *saves them in a external csv file*
- **Restore History** restores all the entries in the Alert Table that have been saved in the csv file

N-MON Fleet Monitor

Active Vehicles
4

Jamming Events
5

System Status
OPERATIONAL
↑ CONNECTED

1 new jamming event(s) detected!

Live Alerts Feed

Clear History

Filters

Map

Restore History

	Vehicle ID	Time	Jamming Type	Jamming Level	Latitude	Longitude
0	TR489GF	2026-05-17 18:15:26	LN	High	45.058000	7.696492
1	AB123CD	2026-05-17 18:15:27	LWF	MID	47.162100	7.662200
2	TR489GF	2026-05-17 18:14:56	TRI	High	45.104061	7.687521
3	DB756JH	2026-05-17 18:15:25	TRI	High	45.043180	7.632285
4	EF345AD	2026-05-17 18:15:25	TRIW	Mid	45.021959	7.665091

18:18

LTE 100

N-MON · DRIVER TERMINAL

INSERT PLATE

AB123CD

CONNECT & MONITOR

MONITORING · AB123CD

SYSTEM · ACTIVE

18:18

LTE 100

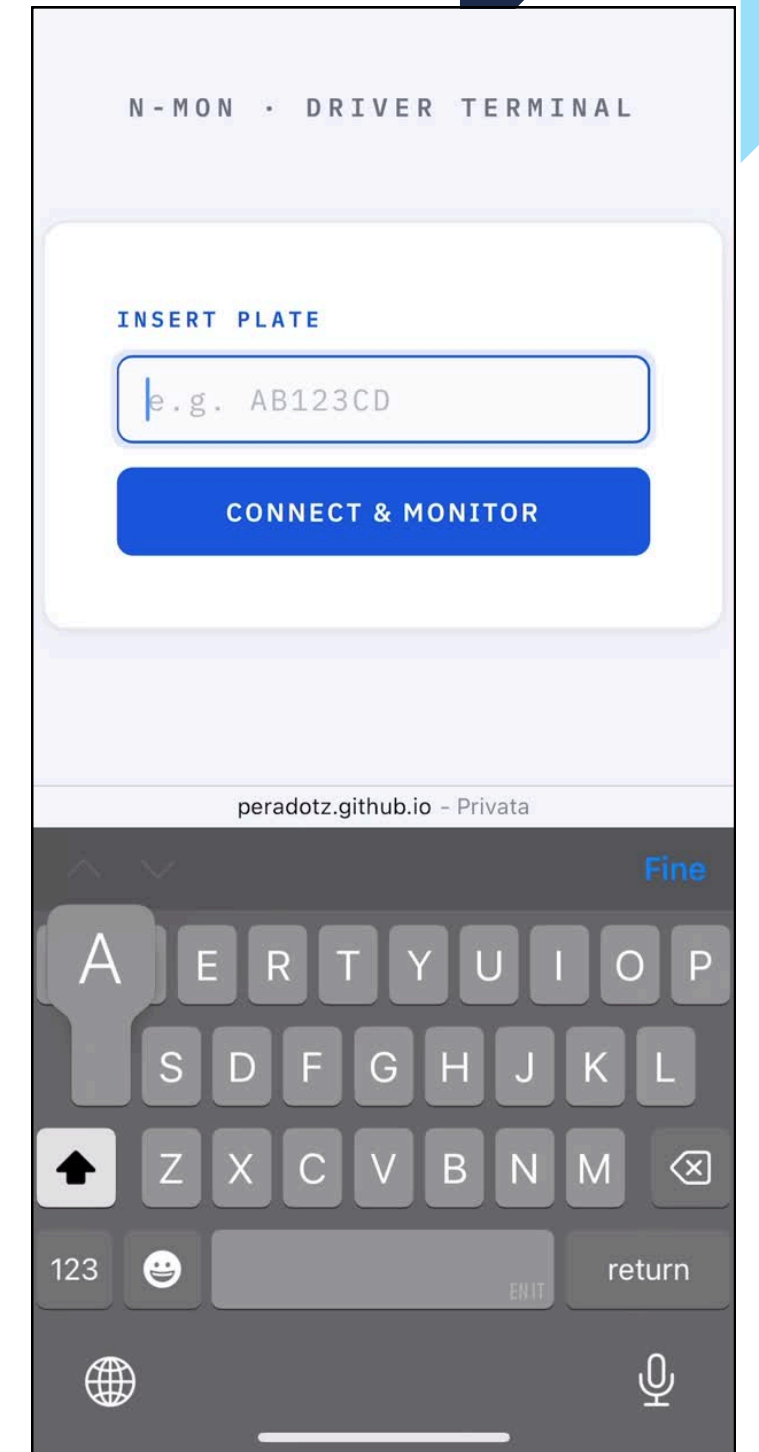
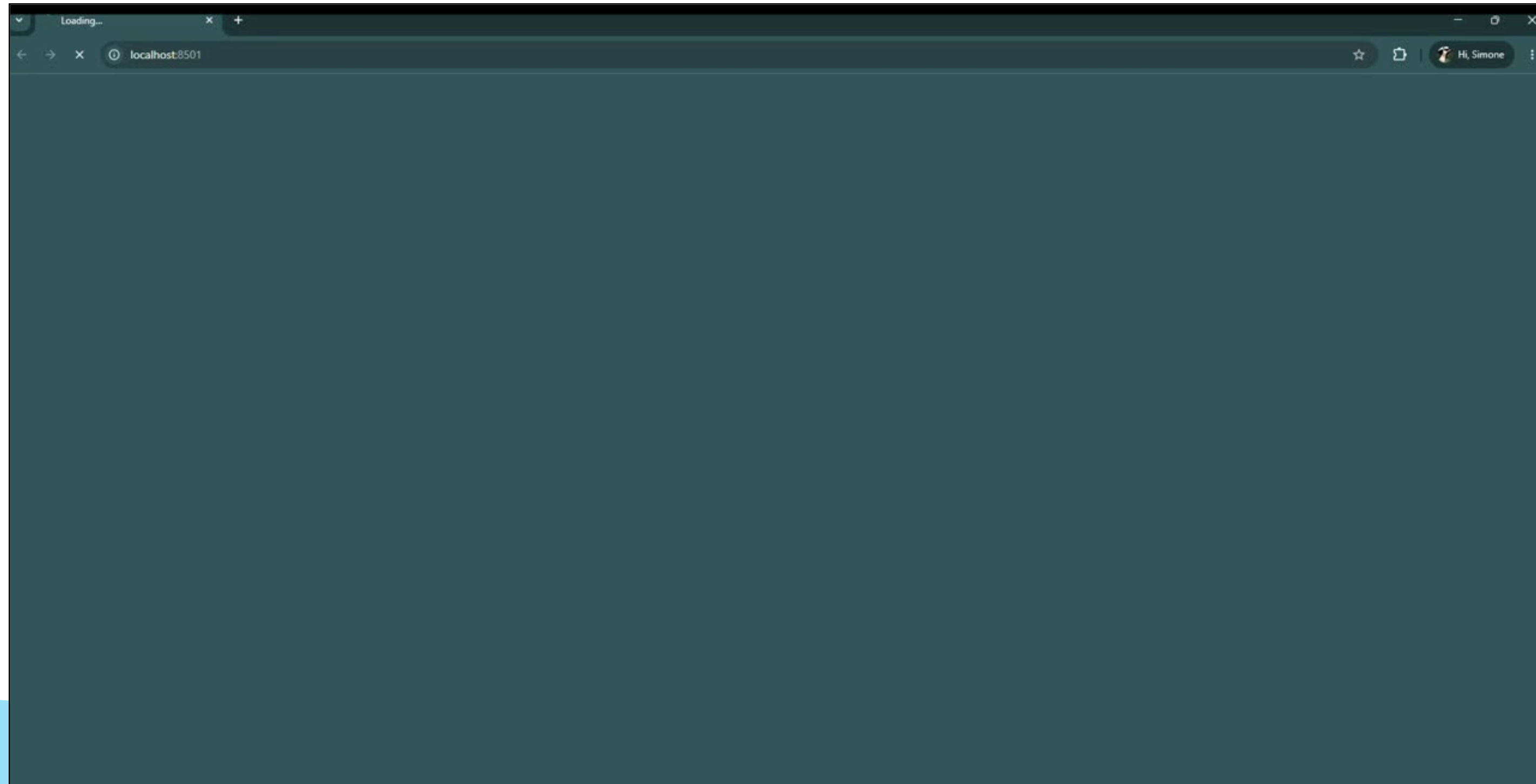
N-MON · DRIVER TERMINAL

JAMMING DETECTED

Vehicle · AB123CD

ACKNOWLEDGE

USER INTERFACE LAYOUT



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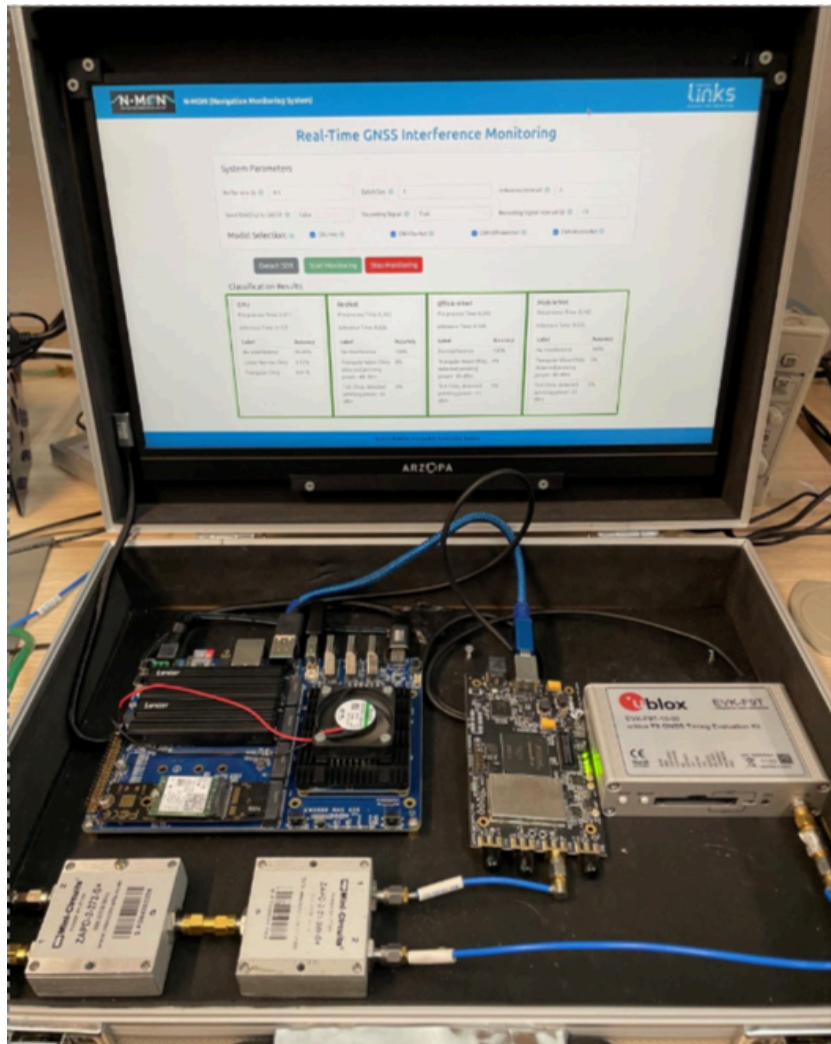
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Link accessible with Polito credentials :

https://politoit-my.sharepoint.com/:x:/g/personal/s343420_studenti_polito_it/EVHoF8byxhZOuSA_gHzWz6ABPUtiU81AzwQXv3SZYcAgyA?e=vecgTL

SATISFIED REQUIREMENTS

TREQ_8

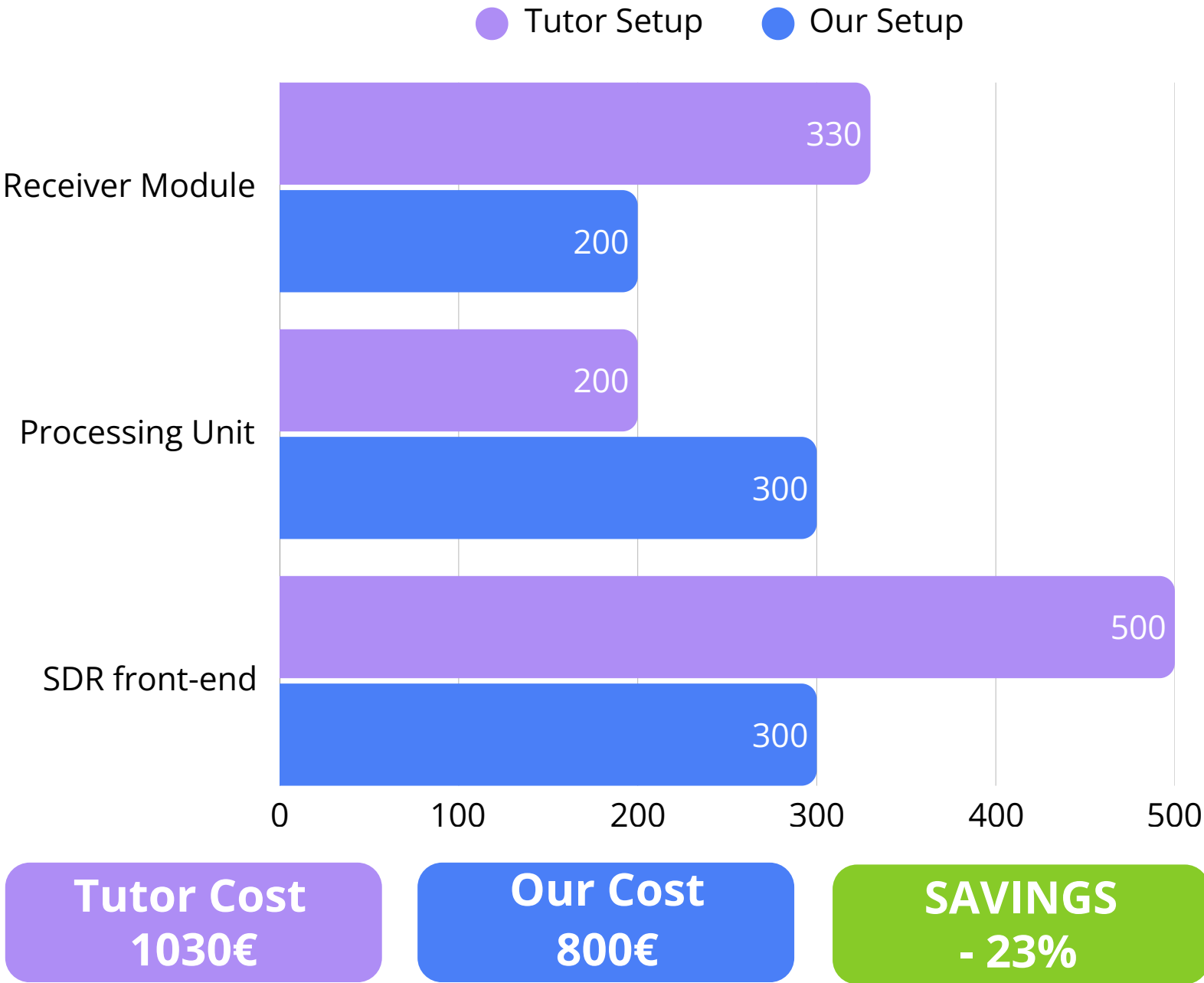


40 x 30 x 8



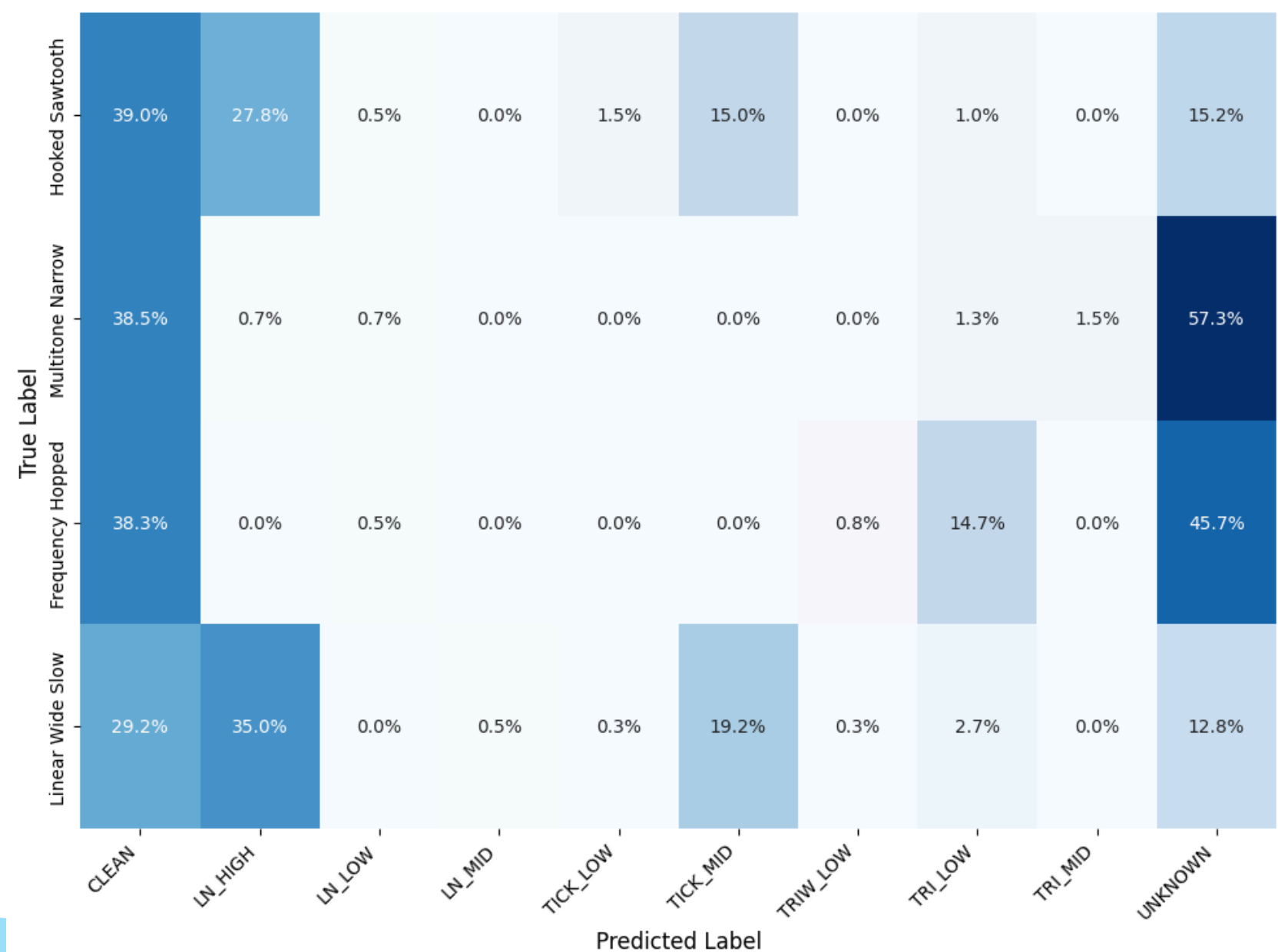
External Box: 23 x 16 x 12
Internal Box: 17 x 13 x 11

TREQ_9



FUTURE IMPROVEMENTS

Confusion Matrix - UNKNOWN Analysis



Open-Set Recognition

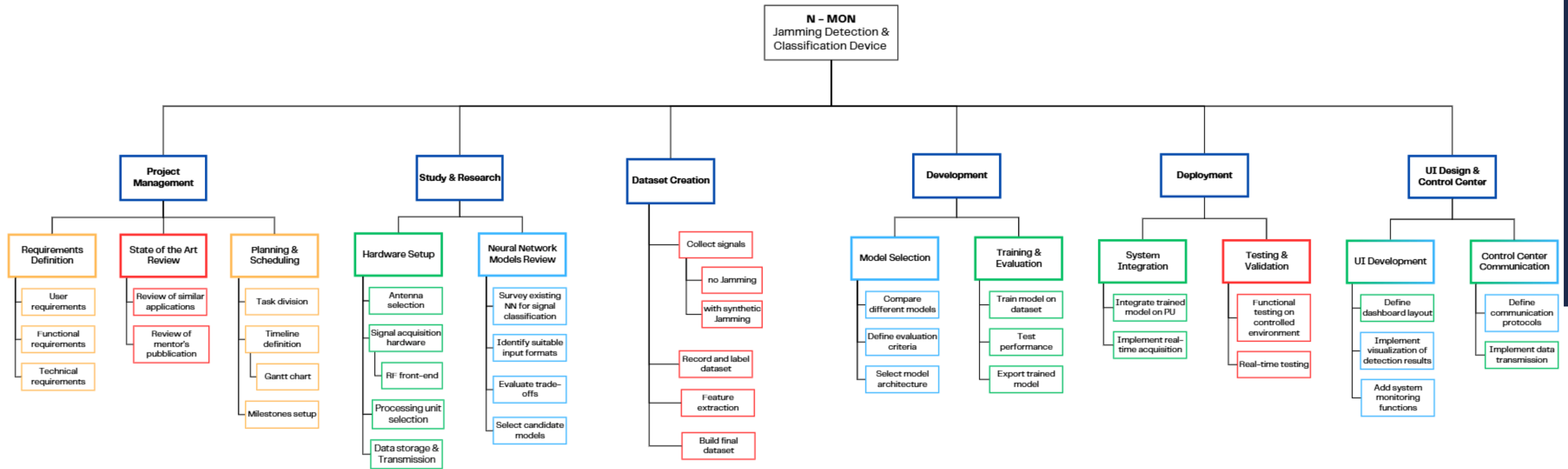
- Current unknown detection rate: 32.8% → Target: >80%
- Unknown jammers fall near known class centroids in feature space (especially CLEAN and TICK_LOW), causing false acceptance
- The baseline model (no OSR) already detects 78% of unknown jammers as some form of jamming activity
- OSR adds a rejection mechanism on top, but currently only improves unknown flagging to 32.8% due to poor geometric separation in the feature space
- Next steps: per-class threshold recalibration and exploration of more advanced OSR methods

Performance Optimization

- Current pipeline latency: 350ms → Target: <100ms
- Bottleneck is CPU-bound preprocessing (81.7% of total time) not CNN inference (9.2%)
- Possible solutions: optimize preprocessing with C++ or ARM-optimized libraries

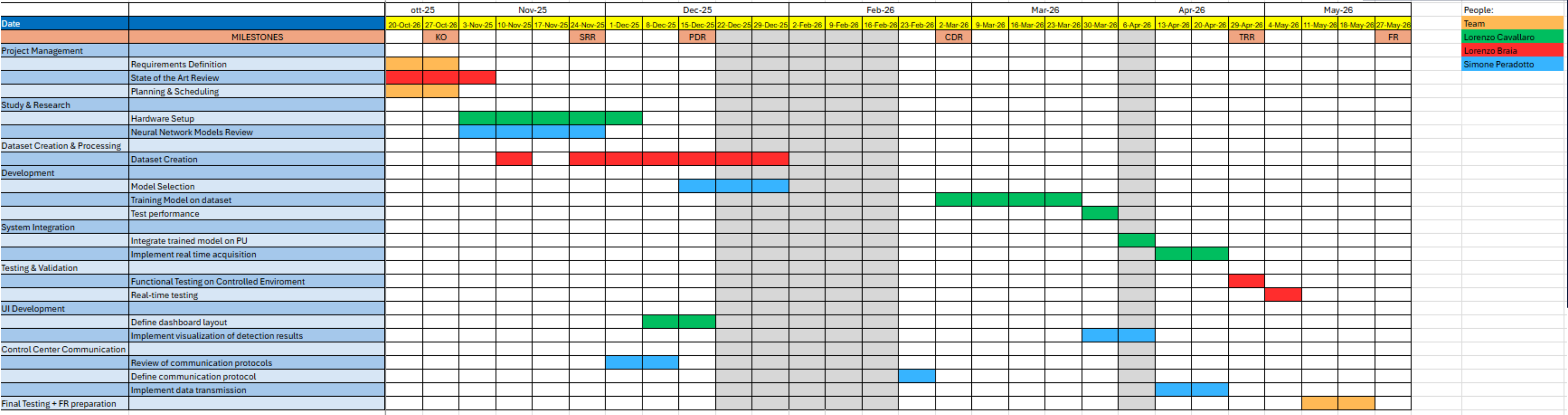
All codes available [here](#)

WBS



<https://www.canva.com/design/DAG14EVzFeY/AZdbRfanr0YUQmWhZWHVmg/edit>

GANTT CHART



Link accessible with Polito credentials :

https://politoit-my.sharepoint.com/:x:/g/personal/s343420_studentsi_polito_it/EVHoF8byxhZOuSA_gHzWz6ABPUtiU81AzwQXv3SZYcAgyA?e=vecgTL



THANK YOU!

For your attention